CHAPTER 11

Programming for the Internet

Finally, we come to what can probably be appropriately described as the crux of the entire book. It is also the final chapter of the book; everything that has come before has been leading up to this. PHP was originally created by Rasmus Lerdorf for processing hypertext (HTML) on web servers (aka "the backend"). It was only natural that PHP's feature set grew to include tools for dealing with HTML-adjacent tasks, such as interacting with the protocols for transferring hypertext (HTTP), connecting to databases, transferring files (FTP), or sending email (SMTP). In the very first chapter – "Getting Started" – we saw how PHP can be run via various server APIs (SAPIs) the majority of which involve some form of web server. PHP even has a limited server built in, though it exists purely for development purposes and is not to be used in any other context. PHP – standing first for *Personal HomePage*, then later *PHP Hypertext Preprocessor* – and the internet have grown together over the last twenty-plus years, and certainly the PHP of today (version 8.4) is a far cry from anything that came before version 5.

The vast majority of code examples in this book have been deliberately created to be agnostic of the SAPI, you should have had equal success typing them into an interactive PHP shell (via `php -a`), running them from files (via `php file.php`), or via the slightly more convoluted web-server SAPIs. The intent was to demonstrate the language features free from context: numbers, strings, arrays, objects… these will all do the same things whether the SAPI is `cli`, `php-fpm`, `cli-server`, and so on. However, there are certain aspects of the language (e.g., some of the string handling functions) that I've kept back until now because they exist to solve common problems when your code executes within the context of an HTTP request. With that in mind, every code example in this chapter is intended to be run via some form of PHP web SAPI: please revisit Chapter 1, "Getting Started", for information on how to get PHP running via a web server if you aren't sure how to do this.

© Paul Tregoing 2025
P. Tregoing, *Exploring PHP 8*, https://doi.org/10.1007/979-8-8688-2136-3_11